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A Mixed Reality Framework for the Assembly
of Irregular Tree Log Structures



ABSTRACT

Timber De-Standardized is a framework that salvages irregular and regular shaped tree logs by utilizing a mixed reality (MR) interface for the design, fabrication, and assembly of a structurally viable tree log assembly. The process engages users through a direct, hands-on design approach to iteratively modify and design irregular geometry at full scale within an immersive MR environment without altering the original material. A digital archive of 3D scanned logs are the building elements from which users, designing in the MR environment, can digitally harvest (through slicing) and place the elements into a digitally constructed whole. The constructed whole is structurally analyzed and optimized through recursive feedback loops to preserve the users' predetermined design. This iterative toggling between the physical and virtual emancipates the use of irregular tree log structures while informing and prioritizing users' design intent. To test this approach, a scaled prototype was developed and fabricated in MR.

- 1 Left to Right: Physical prototype at 1:1 scale, digital model, and structural feedback model visualized with mixed reality tools

By creating a framework that links a holographic digital design to a physical catalog of material, the interactive workflow provides greater design agency to users as co-creators in processing material parts. This participation enables users to have a direct impact on the design of discretized tree logs that would otherwise have been discarded in standardized manufacturing. This paper presents an approach in which complex tree log structures can be made without the use of robotic fabrication tools. This workflow opens new opportunities for design in which users can freely configure structures with non-standardized elements within an intuitive MR environment.

INTRODUCTION

Environmental Observation

Presently active forest management and responsible harvesting practices have been employed to reduce the carbon footprint of timber, one of the most common and sustainable materials for building construction. However, the manufacturing and production process of timber products can affect and pollute its surrounding environments and ecosystems (Adhikari and Ozarska 2018) such as sawmilling industries, the use and production of glue toxins, the process of transportation, etc. Furthermore, the manufacturing process to produce dimensional lumber and plywood products requires the standardization of logs into straight members (Belly et al. 2019). These manufacturing processes also create enormous amounts of waste in the form of tree branches, off-cuts, sawdust, bark, and juvenile trees. As Adhikari and Ozarska (2018) reported in Nigeria: "Out of every 1 m³ of tree that is cut and removed from the forest, about 50% goes to waste in the form of damaged residuals, followed by abandoned logs (3.75%), stumps (10%), branches (33.75%), and butt trimmings (2.5%)."

The waste produced from tree harvesting is primarily comprised of irregular shaped logs and branches that cannot be standardized for manufacturing and consequently cannot be used for standardized applications. Therefore, the use of non-standard tree geometry has the potential to mitigate the carbon impact of industrial dimensional lumber manufacturing processes by salvaging the wood material that is presently considered to be waste. Though the irregularity of the waste produced from tree harvesting is incompatible with standard applications, the irregularity of the wasted tree geometry is easily workable in 3D user interfaces of augmented and mixed reality platforms.

A Framework

Timber De-Standardized is a framework that demonstrates how non-standard materials in the form of irregular tree logs can be salvaged and repurposed to create a tree log assembly through the integration of 3D scanning, mixed reality, and several iterative stages of structural analysis and optimization. Designing with non-standard materials is fundamentally more complex than designing with standard materials because each element is discrete. However, designing in virtual and augmented realities (3D user interfaces) does not privilege one geometry over another; each element is equally compliant because the interaction between geometries is strictly in accordance with the user's intuitive gestures. The intuitive familiarity of working in a 3D user interface (UI), as compared to a 2D UI, has the opportunity of making the UI more accessible to a broader

audience of non-expert and expert users alike. Salvaging non-standard tree logs necessitates the cataloging of such elements with a 3D scanner in order to link the digital mesh derived from the 3D scan with each discrete physical member. *Timber De-Standardized* creates a platform that utilizes what would otherwise be considered as discarded tree logs to democratize design through the employment of a mixed reality environment. Mixed Reality (MR) enables the interaction between human, computer, and one's immediate place through immersion of both physical and digital interfaces (Microsoft 2020). In order to test the framework, a MR-designed spatial tree log assembly is analyzed and optimized to reduce structural deformation as a scaled prototype that was originally proposed for a conceptual housing project. The MR environment initiates the use of virtual instructions for the fabrication and assembly of the prototype. For the purpose of contextualizing and testing this framework, the research goals were to (1) establish design and structural parameters specific to working with irregular tree logs, (2) develop an interactive workflow for users to manipulate the tree logs for design and fabrication between the mixed-reality, digital, and physical environments, and (3) to develop an optimization process that would prioritize the users' design intent. *Timber De-Standardized* is a framework that utilizes MR to initiate a platform that aims to democratize design agency through several iterative design and structural analysis stages followed by the provision of direct fabrication and assembly instruction, all of which are linked to a physical catalog of salvaged non-standard tree logs.

STATE OF THE ART

Digital Log Constructions

Recent projects such as the *Wood Chip Barn* (Self et al. 2016), *Limb* (Von Buelow et al. 2018), *Branch Formations* (Alner et al. 2021), and *torinosu* (Eri Sumitomo Architects 2020) explore the potential of repurposing irregular tree geometries for structural applications in architecture. *Wood Chip Barn* at Hooke Park was revelatory in its workflow between scanning trees, selecting trees, and indexing 'Y' shaped members to design a truss with selected members, structurally evaluating these members, and finally to fabricating a *Tree Fork Truss* (Mollica and Self 2016). *Limb* continued the exploration of the tree bifurcations as it relates to shell structures, (Von Buelow et al. 2018), while *Branch Formations* also studies similar tree bifurcations, this time with regard to structures derived from a tetrahedral based recursive aggregation algorithm (Allner et al. 2019). Because of the irregularity of the non-standard members selected for design, each of these three projects relies on robotic fabrication for the construction of their final assemblies. Still reliant on the

process of 3D scanning, Aki Hamada and Eri Sumitomo led a team on the repurposing of irregular tree logs into a reciprocal frame in *torinosu* (Eri Sumitomo Architects 2020). Instead of relying on robotic fabrication, the team utilized an augmented reality interface for the manual fabrication and assembly of *torinosu*. Augmented reality (AR) is a composite view of the real world and a virtual scene that is digitally generated for the perspective of the user (Nakata et al. 2012). In recent years, AR has become an increasingly popular tool for architectural practices to project virtual instruction into a real environment (Song et al. 2021).

A Familiar Environment

AR has become an increasingly popular area of research in architectural based design practices (Song et al. 2021), as it has shown to enhance education and increase student participation (Arici et al. 2019). AR offers a number of affordances as a user interface (Bowman et al. 2006) and as a tool for digital construction (Abe et al. 2017). As a user interface (UI), there is a greater degree of 'unnatural' interaction through a 2D interface as opposed to a 3D interface when orienting, transforming, or otherwise manipulating 3D information (Alaikseyeu et al. 2012). In other words, 3D information can be more fully represented and communicated through a 3D UI (Alaikseyeu et al. 2012). With AR, virtual models made digitally can be displayed and seen in physical space through the live view of a computer device (Nakata et al. 2012). Similar to AR, MR is the intersection between virtual and physical realities with which a haptic stimulus is also presented (Skarbez et al. 2021). This becomes a powerful design tool as it gives the user control of how digital models are placed, oriented, and perceived with respect to the physicality of their environment.

These tools create new approaches in design and fabrication where, what would otherwise be considered complex digital models, can be made without robotic-based fabrication techniques through the instruction provided in a 3D UI. This concept was explored in the *Woven Steel Pavilion* (2018) (Jahn et al. 2019) during a workshop at the 2018 CAADRIA conference in Beijing, in which researchers from Fologram tested their mixed reality designed interface as a tool for metal rods bending instruction. The users were able to accurately bend each rod in several multi-axial directions as informed by the MR environment (Jan et al. 2019). As inferred through this project and other projects such as *torinosu*, MR becomes highly relevant in the manipulation, reorientation, and communication of the multi-axial nature of irregular tree logs through the 3D UI visual instruction.

Through the hybridity of human input and MR technologies,

design can become an iterative and generative tool for engaged users. When informed by immediate structural simulations, the processes of fabrication and assembly can become a localized educational instruction manual for the users. With the added stages of structural simulation to a MR environment, *Timber De-Standardized* enables a participatory and iterative approach by providing the user with immediate structural feedback throughout the design process. The further active participation in such a framework enhances the design agency of the participant from user to co-creator of custom non-standard log components. The spatialization and tangibility of an AR environment (similar to the proposed MR setup) empowers users to interact with 3D information through an authentic learning environment (Cheng and Tsai 2012).

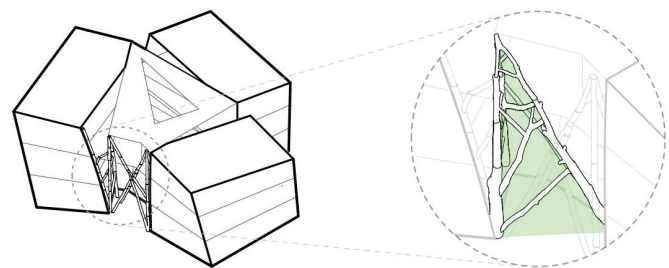
METHODS

Design Approach

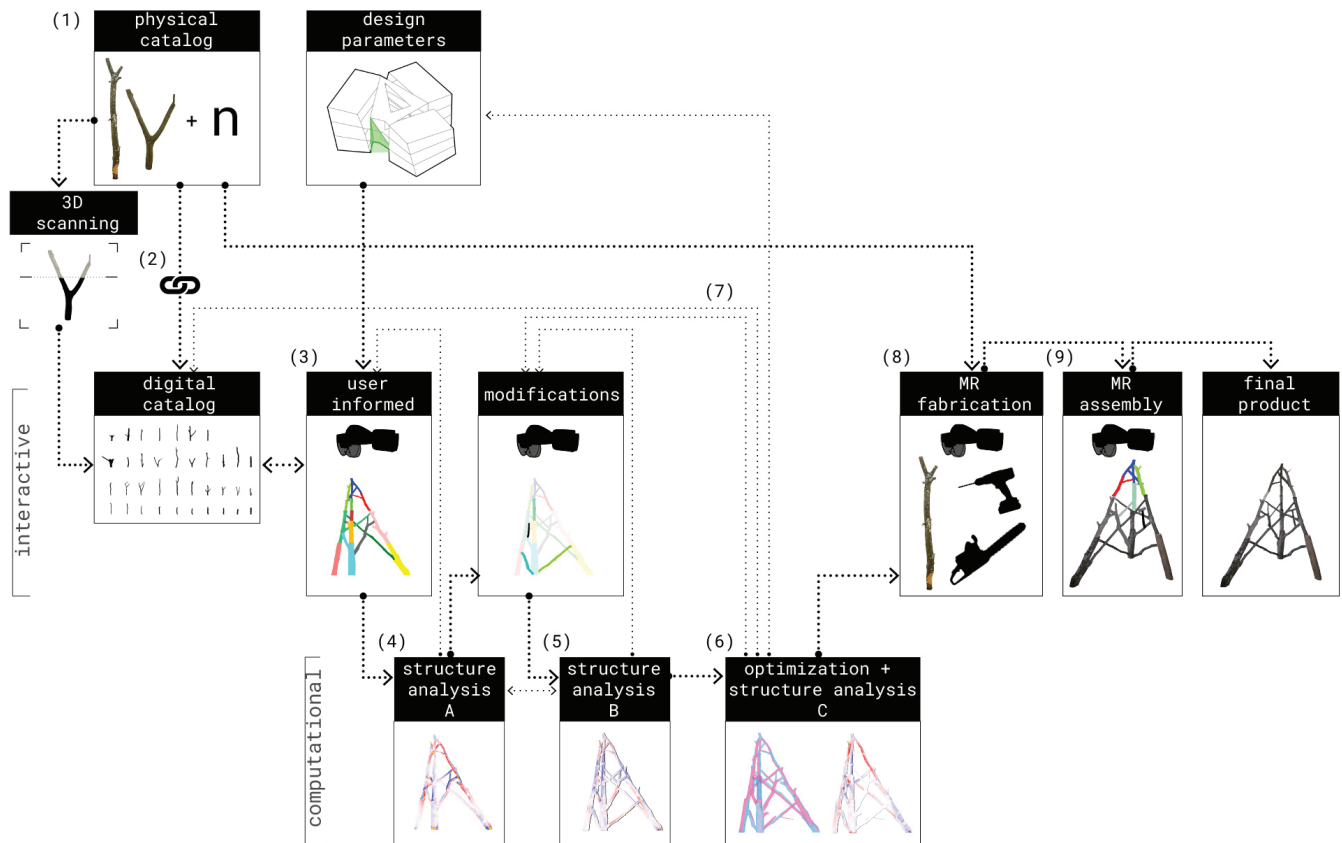
The prototype is part of a three-story housing complex that explores innovative uses of ash wood. The housing complex is composed of three twisted rectangular volumes that enclose a central courtyard. The voids created from the corners of the twisted volumes infer the bounding geometry from which the tree log assembly is derived. The voids also serve as the continuation of the adjacent circulation corridors and facilitate the movement of the inhabitants across the complex. The prototype for *Timber De-Standardized* is a portion of one of the voids at the entry to the housing complex (Figure 2). The volume is a triangular pyramid with a circulation corridor at each level. In this conceptual housing complex, each level of the structure is laterally supported with fixed connections at the ground level. However, for the purpose of the prototype, the structure is self-supported and the base connections are not fixed. These aspects of the housing complex are the driving geometry that informs the tree log assembly designed in MR from irregularly shaped logs.

Workflow

Timber De-Standardized is a framework that enables an iterative process of design through several recursive



2 Housing complex with structure

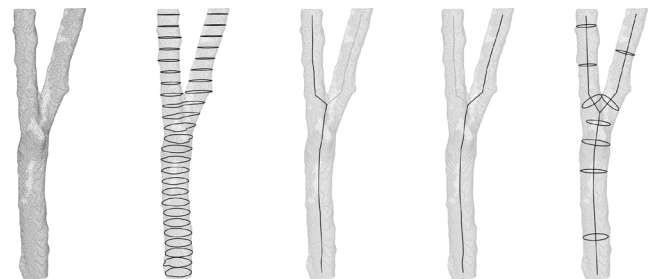


3 Workflow diagram

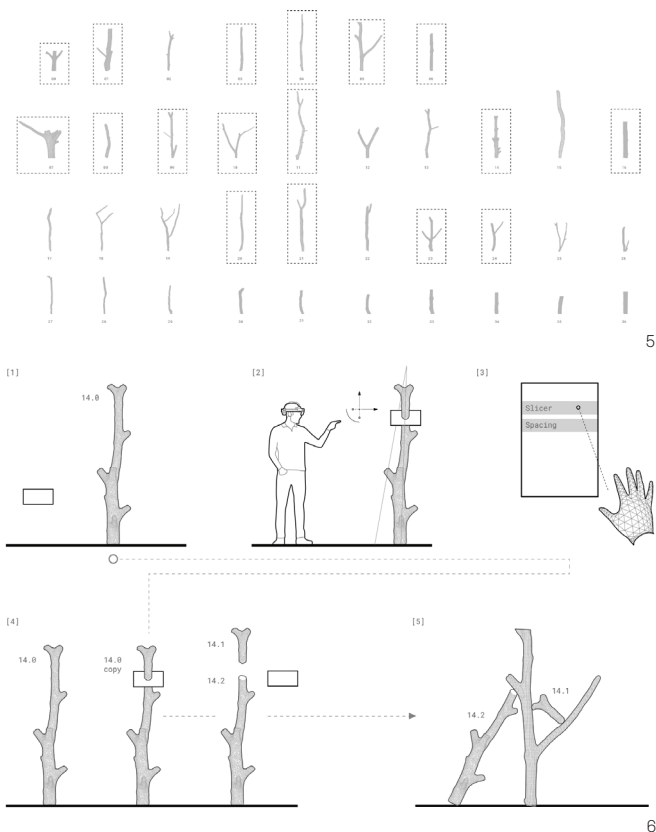
stages of structural analysis in a MR environment (Figure 3). For the purpose of this prototype and the demonstration of this framework, irregular tree logs were (1) sourced from a local groundskeeping entity, and (2) cataloged and indexed using a 3D scanner. Next, the tree logs were used as the (3) 'building blocks' to design a tree log assembly in the MR environment. The resulting structure was immediately (4) analyzed for its structural viability as the first structural loop. From this, (5) the second structural loop, in which the user can continue to modify the design in MR according to the results observed from the structural analysis feedback loop. Once the user completes the modifications, (6) the third structural analysis 'jitters' the model to find the optimum orientation and alignment of the members placed by the user, then (7) the resulting structure is re-indexed. The automatic re-indexing (8) instructs the user on how to proceed with the following fabrication steps (cutting the members and drilling holes for their joinery) in the MR environment. Finally, (9) the MR environment continues to be utilized as each member is assembled. The final structure is the derivative of an iterative process of MR design, structural simulations and a low-tech fabrication and assembly process enabled by the MR environment. This framework serves as a participatory and educational platform for the design and construction of structures sourced from irregular tree logs.

Cataloging

As previously mentioned, irregular, discarded tree logs were sourced from an already existing local stockpile. A wide variety of profiles from the stockpile were selected and scanned using Scanner - Structure SDK, a mobile application for iPad and iPhone by Occipital, Inc. with the associated accessory hardware, Structure. Each tree log was indexed independently as a digital mesh. The centerlines from each tree log (Figure 4) is derived by first contouring the log to get a series of cross sections that were connected together at their center. In the case of tree logs with one or more bifurcations, duplicate lines were removed. Each log, its centerlines, and respective cross sections were cataloged with a unique file name, so that any one or several files could be referenced into a single library file (Figure 5). Using Fologram, a mixed



4 Computational analysis of individual log



5

physical space. Once the digital tree meshes are positioned in the MR environment, the user can use the 'slicer,' a plane for slicing the digital tree log meshes, to determine the cut locations for each log. In this process, after each 'slice' operation, a copy of the whole tree log mesh is created in order to re-index the part to the whole during the fabrication process. This immersive process engages the user work iteratively according to their own intent and design exploration (Figure 6). The user can modify and make unique parts without altering any of the discretized physical tree logs until the final design is set.

When designing in the MR environment, it was necessary to illustrate the design parameters of one of the voids from the conceptual housing project to guide the user in their design. As such, floor slabs, structural guides, and a corridor were visibly represented and served as the guiding principles for the user (Figure 7).

Structural Feedback Loop_A (Real-Time Visualization)

The structure was designed in Fologram with HoloLens according to the design parameters defined by the user. However, this method of working does not presently allow for the centerlines of the mesh to automatically snap together. In order to test the structural viability of the augmented reality designed structure, each tree mesh was evaluated as shell structures, while the assumed dowel joint connections between each tree mesh were evaluated as beam elements using Karamba 3D (Preisinger and Heimrath 2014), a parametric engineering plugin for Grasshopper 3D and Rhino 3D. The location of the dowel joint connection was determined by both the size and location of the 'slicer'. In the first structural loop, Structure Loop_A, the user can see the structural parameters as well as deformation and utilization in the Fologram environment.

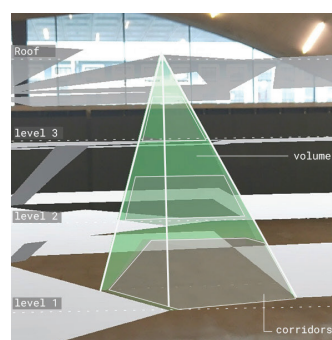
Structural Feedback Loop_B (User Modification)

Based upon the information visually communicated to the user, additional members can be manually added by the

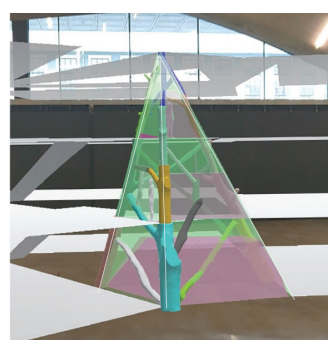
reality software for mobile devices and Rhino 3D and Grasshopper 3D (digital modeling softwares by Robert McNeel & Associates), with HoloLens, a mixed reality pair of smart glasses by Microsoft, the library file is the platform through which the user can select tree branches from the digital catalog to modify and manipulate within the MR environment.

Mixed Reality Design

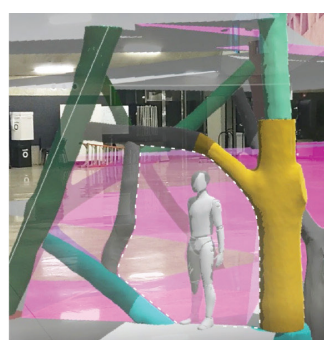
The Fologram environment with HoloLens easily allows the user to move, orient, and position the scanned logs in space as desired. This enables the user to design and prototype at scale, while giving the user an accurate sense for the actual size of the building components in relation to



Parameters informing the placement of log members



Log selection and placement according to parameters



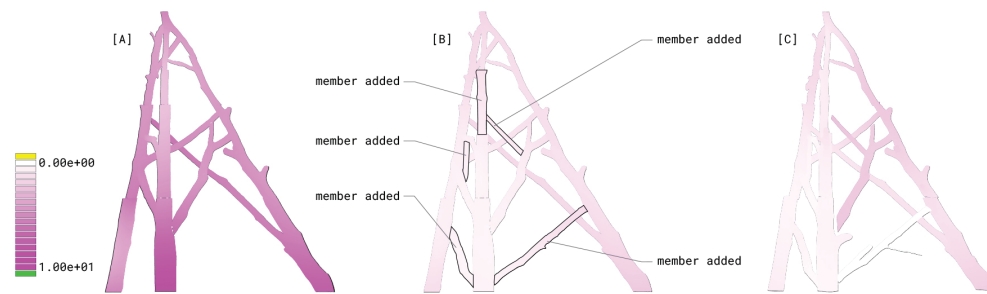
Passage entry framed by logs

5 Whole log catalog

6 Mixed reality interactive workflow diagram

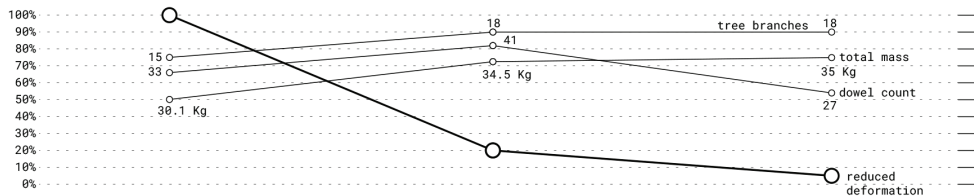
7 Initial bounding parameters

7

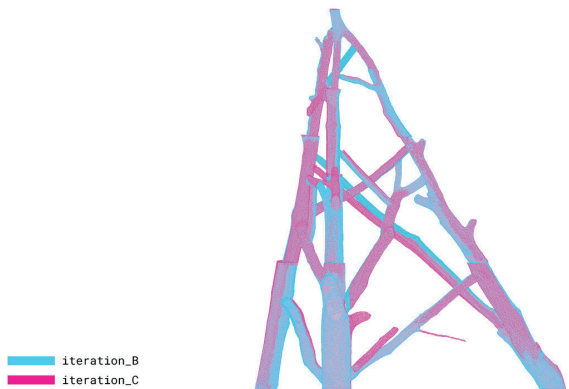
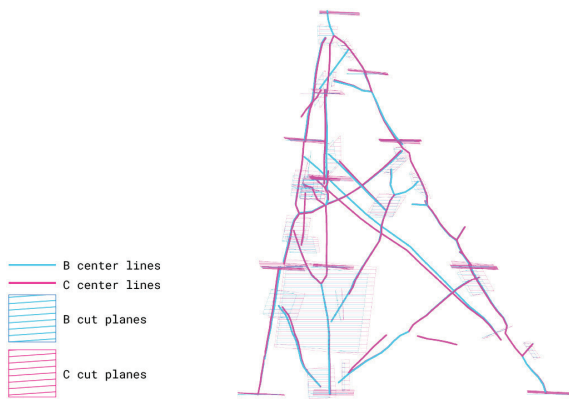


8 Resulting structural analysis at Structure Loop_A, B, & C

9 Structure Loop_C



8



9

user using the MR environment to reduce the total deformation of the model. This is the second structural loop (Structure Loop_B). In this prototype, the user added five members and removed two from the previous iteration. Figure 8 illustrates the reduction of the deformation between Structure Loops A, B, and C.

Structural Feedback Loop_C (Realignment)

Upon completing the second iteration of design, the

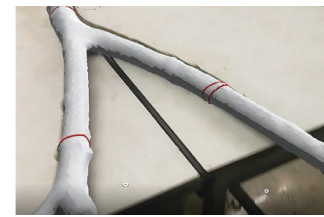
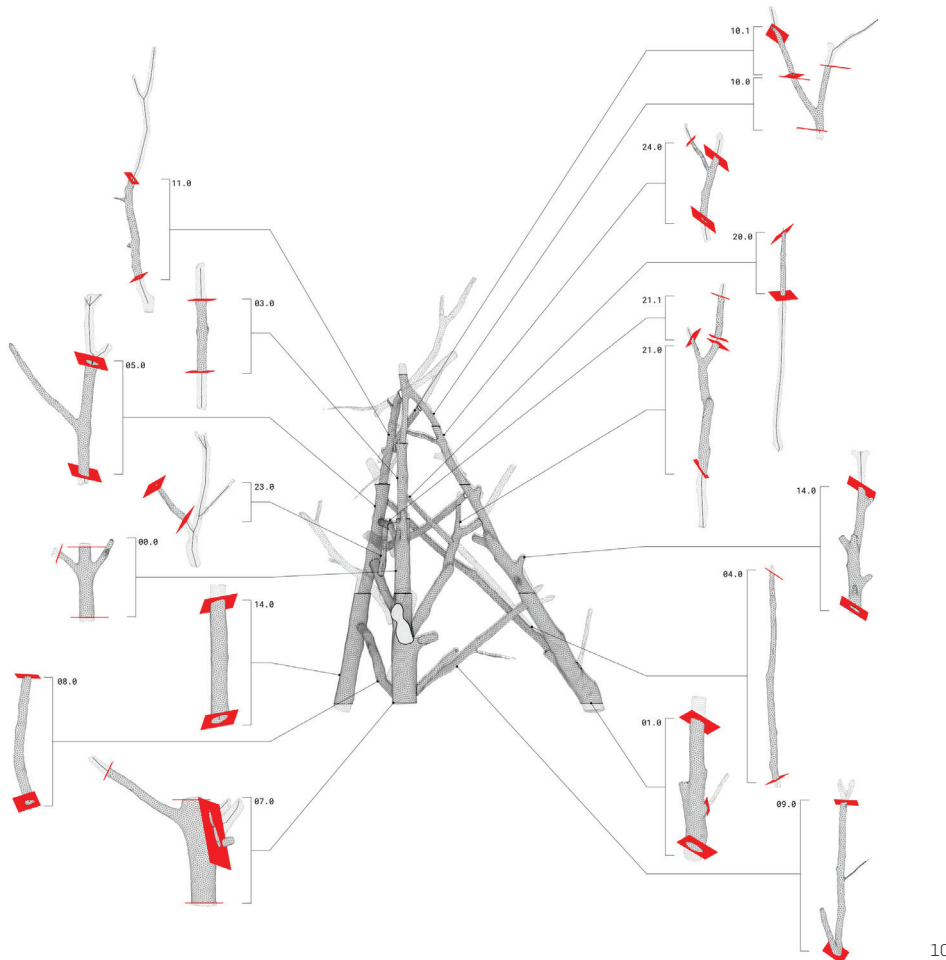
geometry from the model is then optimized with Galapagos, an evolutionary solver for Grasshopper 3D, by realigning and rotating each member around its local x, y, and z axes to reduce the total deformation of the model. Additionally, the 'slicers' used to split the digital tree meshes in the MR environment are able to transition and reorient in their local x, y, z axes. Each connection derived from the 'slicer' is referenced as a beam element in the Karamba 3D model. Structure Loop_C utilizes the Galapagos solver to adjust variables with the goal of reducing the total deformation in the model. Figure 9 illustrates the differences between Structure Loop_B and Structure Loop_C.

Re-Indexing

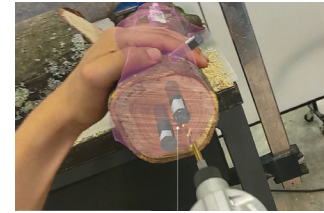
The final transformation of each tree mesh is recorded and applied to each of the elements associated with the tree mesh, i.e. center lines, cross sections, layer, and object attributes, etc. This method allows for the reorientation of each part to its original whole according to the orientation of the derivative optimization before the parts were split by the 'slicer.' The members are also referenced to the associated members in the physical catalog. This allows the user to see the updated cut location for each tree branch as well as the location to drill the holes after each tree branch is cut (Figure 10).

Fabrication

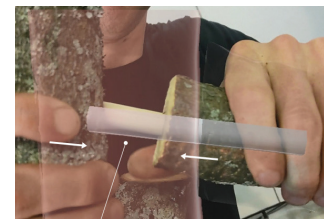
The fabrication for each individual log required strategic planning and the use of standard power tools as guided by the HoloLens. In order to align the digital tree log meshes with their respective physical tree log, it was crucial to orient the digital tree log mesh to the natural resting position of the physical tree log in Rhino 3D.



11



12



13

- 10 Re-indexing
- 11 Digital log simulated in natural resting position on a flat surface to overlay with the physical log
- 12 Fabrication of joint locations
- 13 Dowelled joints connecting two members

For this task, Kangaroo (Piker 2013), a built-in physics simulation plugin for Grasshopper 3D, was utilized to simulate the logs' natural resting position on a flat surface, which could then be viewed in the MR environment. One by one, the digital logs were positioned on a table and were then overlaid with the physical logs until both profiles matched (Figure 11). With the assistance of the MR environment, section cuts were visibly displayed on the logs. For safety purposes it was decided to first mark and trace these section cuts with a marker and then have the logs cut with a chainsaw. The same process would be followed for the location of the joints. However, because the joints used for this prototype are dowelled connections made from dowel rods, there were instances where the connections between logs occurred at different angles. To accommodate this, the MR environment was used to visualize the location and angle in which the drill bit had to enter (Figure 12).

Assembly

After pre-drilling each hole for the dowel-based connections, the large, upright log at the base was selected to be the first member to guide the assembly of the rest of the members. The prototype uses dowelled joints (Figure 14)

with each member being either an end-to-end connection or a side-to-end connection. Two different dowel dimensions are used in the prototype. Most members use two dowels per connection to mitigate any type of local x or y rotation. The wearer of the HoloLens instructed the other two users as to the assembly of the members and often directly positioned the members to ensure the angles of each member were accurate to the digital model (Figure 15). The use of a MR environment was necessary because of the complex, multi-axis nature of each non-standard element.

RESULTS AND DISCUSSION

The Final Structure

The built prototype demonstrates the possibilities of working and designing with irregularly shaped logs at the intersection between MR and structural optimization. The prototype serves as a proof of concept highlighting key design decisions made by the user. This is evident in the log selection, the unique log segments created by slicing locations, and the placement of each individual member with the thicker logs placed towards the bottom and the thinner ones placed up top.



14 Prototype assembly

The final prototype is a self-supporting structure, measuring 9 feet in height and covering an area of 11 square feet. The structure uses a total of 16 salvaged tree logs cut into 18 members (Figure 15) with 28 dowel connections. The time required to complete the assembly was seven hours.

Reflections

Throughout the three phases of this project, several findings were made regarding the MR workflow. There are multiple advantages to the interactive design process: the MR technology provides intuitive commands that render the interaction with digital models almost physical. This analog/digital approach towards design enables the user to freely design and physically explore structural designs according to their intuition, without the actual weight of the log. Unlike designing through a 2D UI, the MR environment allows the user to truly understand the scale and context in which the design will be situated. As for the fabrication process, MR provides a platform where irregular shaped logs can be cut and drilled fairly accurately according to the information provided by the 3D UI.

The framework performed well for the purpose of this prototype, however, due to the limitations of the MR interface, the constant switch between HoloLens and the computer (for the purpose of referencing various logs) significantly slowed the design process. Another challenging aspect that emerged from working in MR was the issue of holographic drift, which tended to complicate the alignment and precision between the physical and digital tree logs and the resulting cut lines and drill locations. Advanced drift correction has recently been implemented in Fologram's 2021 sibling program Twinbuild and could be tested in future experiments.

Furthermore, the MR workflow discussed in this paper assumes that users have a basic foundational knowledge in design and construction. Due to the intuitive nature of the

workflow, almost anyone can operate and design within the mixed reality setting. However, someone with the appropriate knowledge and understanding of structure is better suited to work and design with the material.

Future Investigations

The next series of investigations will focus on improving the precision of the joint connections within the mixed reality framework. By building upon the interactive workflow, the joinery-making process could be further developed beyond simple dowel connections.

The scale of the prototype was manageable and parts were easy to assemble. However, moving forward, working with larger logs could present additional logistical challenges during the assembly process, as large machinery will be required to move members into position. The downside to scaling up is that part of the direct user implementation will be relegated to machinery coordination. Moving up in scale presents additional design parameters that were not factored into the structural analysis of this prototype.

CONCLUSION

Key Contributions

The MR workflow realigns environmental, technological, and social aspects within the construction industry. Natural logs that were once considered waste can now be repurposed and utilized for new opportunities in design and construction. Along with these new opportunities is the reduction of the carbon footprint of the manufacturing process. With the use of mixed-reality tools, the design, fabrication, and assembly of complex irregular structures can be accessible without the use of robotic infrastructure. The process engages the user through a direct, hands-on design approach to iteratively manipulate, slice, modify, and assemble their customized non-standard kit-of-parts at full scale within an immersive environment. This approach realigns users' interaction and design agency with a computationally and digitally driven protocol. The uniqueness of the MR workflow is that it preserves the design aesthetic set by the user despite the structural optimization phase. The end result is a structurally viable, spatial, non-standard, log assembly that maintains and facilitates the design intent originally predetermined by the user (Figure 16).

Future Value/Potential

Due to the flexibility and adaptability of the framework, *Timber De-Standardized* can be utilized to design custom structural systems. Furthermore, it presents the construction and environmental viability of using non-standard building material. The approach can be expanded to



15



16

15 Log pieces used

16 Log labeled with centerline to visualize its multi-axis nature

innovate irregular building envelopes made from wood or the use of small round wood elements to create viable conventional structural systems such as trusses or space frames. Additionally, the MR workflow provides an intimate interactive platform between users and the building material. The hybridized workflow takes the physical act of making and combines it with immersive technological tools, to create a computational platform that can be used and engaged by almost anyone. The hands-on workflow within the MR environment provides a new experience, one that gives the user a greater sense of design authorship. Moving forward, there is potential in exploring how the workflow can engage multiple users to coordinate and work together as a team in the MR environment. A team dynamic would restructure and introduce new variables to the current framework.

Timber De-Standardized realigns how we interpret and utilize timber in our current design and construction models. More importantly, it presents a viable minimum infrastructure solution to our environmental crisis regarding the manufacturing and processing of lumber based products.

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